

K. PALPANDI

Electrical & Electronics Engineering Student

+91 70920 68775 | palpandik583223105015@nprcolleges.org | T. Mettupatti, Madurai, Tamil Nadu
[linkedin.com/in/palpandi-k-07ba552a4](https://www.linkedin.com/in/palpandi-k-07ba552a4) | palpandi-portfolio.vercel.app

CAREER OBJECTIVE

Motivated Electrical and Electronics Engineering student seeking to build a career in embedded systems, industrial automation, and power systems. Eager to apply strong technical fundamentals, hands-on project experience, and problem-solving skills to contribute to innovative engineering solutions.

EDUCATION

B.E. — Electrical & Electronics Engineering

Expected 2027

NPR College of Engineering and Technology, Natham

CGPA: 6.88

TECHNICAL SKILLS

PLC Programming • Industrial Automation • Electrical Maintenance • Embedded Systems • Arduino • Circuit Design • Power Systems • MATLAB

INTERNSHIP / WORK EXPERIENCE

Srivari Innovative Automation Solutions, Vadipatti — Bhargave Rubber Pvt. Ltd.

1 Month

Internship Student

- Gained hands-on exposure to industrial machine operation and handling within a manufacturing environment.
- Developed practical knowledge of automation processes and equipment maintenance procedures.

PROJECTS

Smart Energy Meter with Smart Billing System for Real-Time Monitoring, Cloud Analytics, and Automated Slab-Based Billing Using IoT

- Designed an IoT-based energy monitoring system that measures electricity consumption in real time and transmits data to the cloud for monitoring and analysis.
- Implemented automated slab-based billing logic to calculate electricity charges and enable remote tracking of usage and estimated bills.

Technologies: ESP32/Arduino, Current & Voltage Sensors, LCD Display, Wi-Fi, Cloud Platform, Embedded C

Fire Detection and Firefighting Drone/Robot

- Built a firefighting robot and drone system capable of detecting fire and responding with an automated water-pump-based suppression mechanism.
- Integrated sensors and a camera module for real-time fire detection and remote-controlled navigation.

Technologies: Arduino, ESP32, Sensors, Motors, Camera, Water Pump, IoT

CERTIFICATIONS

- PLC Programming
- MATLAB
- Industrial Automation

ACHIEVEMENTS / ACTIVITIES

- National Cadet Corps (NCC) — 'B' and 'C' Certificate holder
- 2nd Prize, Paper Presentation — "Fire Fighter Drone," Sethu Institute of Technology
- Participant, Ampathon National Level Hackathon 2026, Andhra Pradesh
- Participant, Smart India Hackathon 2024 — Project "Walk Bot"
- Presented research paper "Smart Energy Meter with Smart Billing System for Real-Time Monitoring, Cloud Analytics and Automated Slab-Based Billing Using IoT" at ICSJET 2026 (International Conference on Smart Innovation in Engineering and Technology)

SOFT SKILLS

Communication • Teamwork • Problem Solving • Time Management